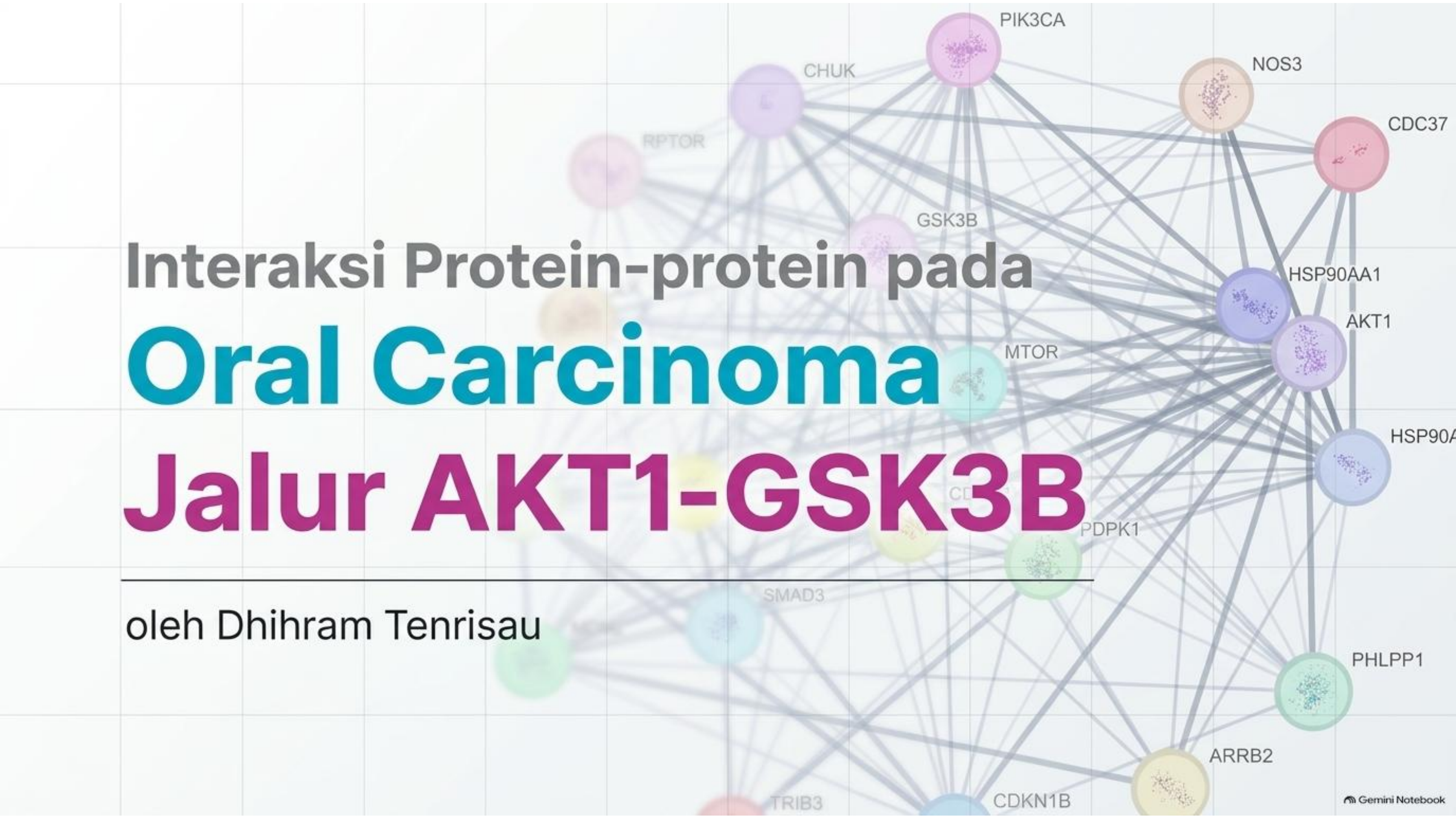


A complex network diagram showing interactions between various proteins. Nodes are represented as colored circles (purple, orange, red, blue, green, yellow, light blue) and are interconnected by a dense web of grey lines. The nodes are labeled with protein names: PIK3CA, CHUK, RPTOR, GSK3B, MTOR, NOS3, CDC37, HSP90AA1, AKT1, HSP90A, PDK1, PHLPP1, ARRB2, CDKN1B, TRIB3, SMAD3, and CD. The title text is overlaid on the left side of the diagram.

Interaksi Protein-protein pada **Oral Carcinoma** **Jalur AKT1-GSK3B**

oleh Dhihram Tenrisau

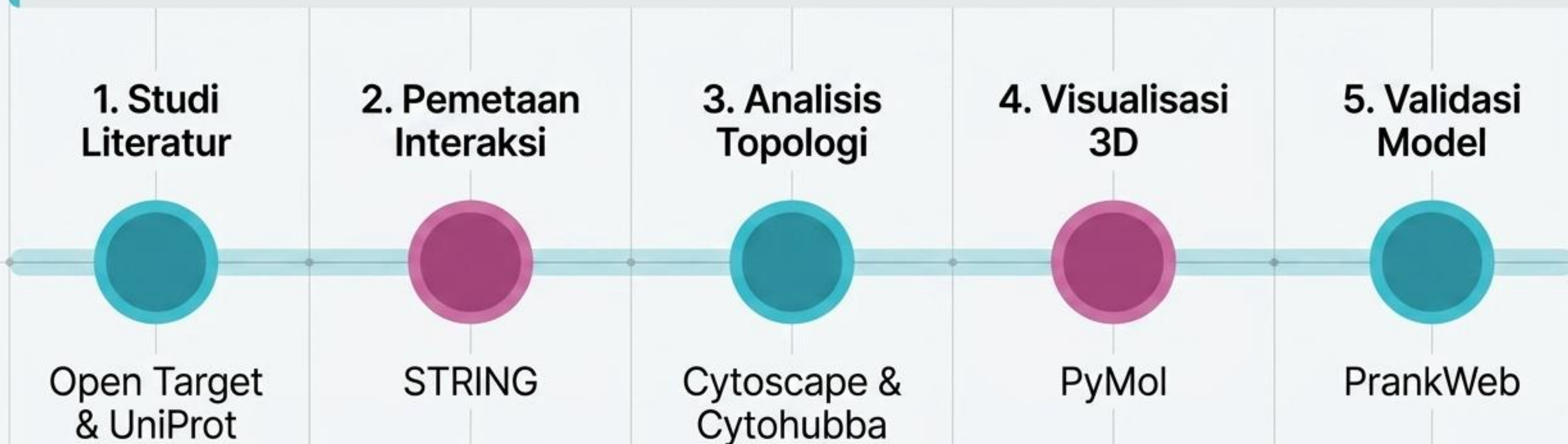


Karsinoma rongga mulut (oral carcinoma) memiliki morbiditas tinggi, didorong oleh disregulasi jalur sinyal seluler PI3K/AKT.





Menganalisis topologi jaringan interaksi protein AKT1-GSK3B serta mengkarakterisasi situs pengikatan dan situs aktif molekuler untuk memahami patogenesis.





Pemetaan STRING & Analisis Topologi

21

Nodes

138

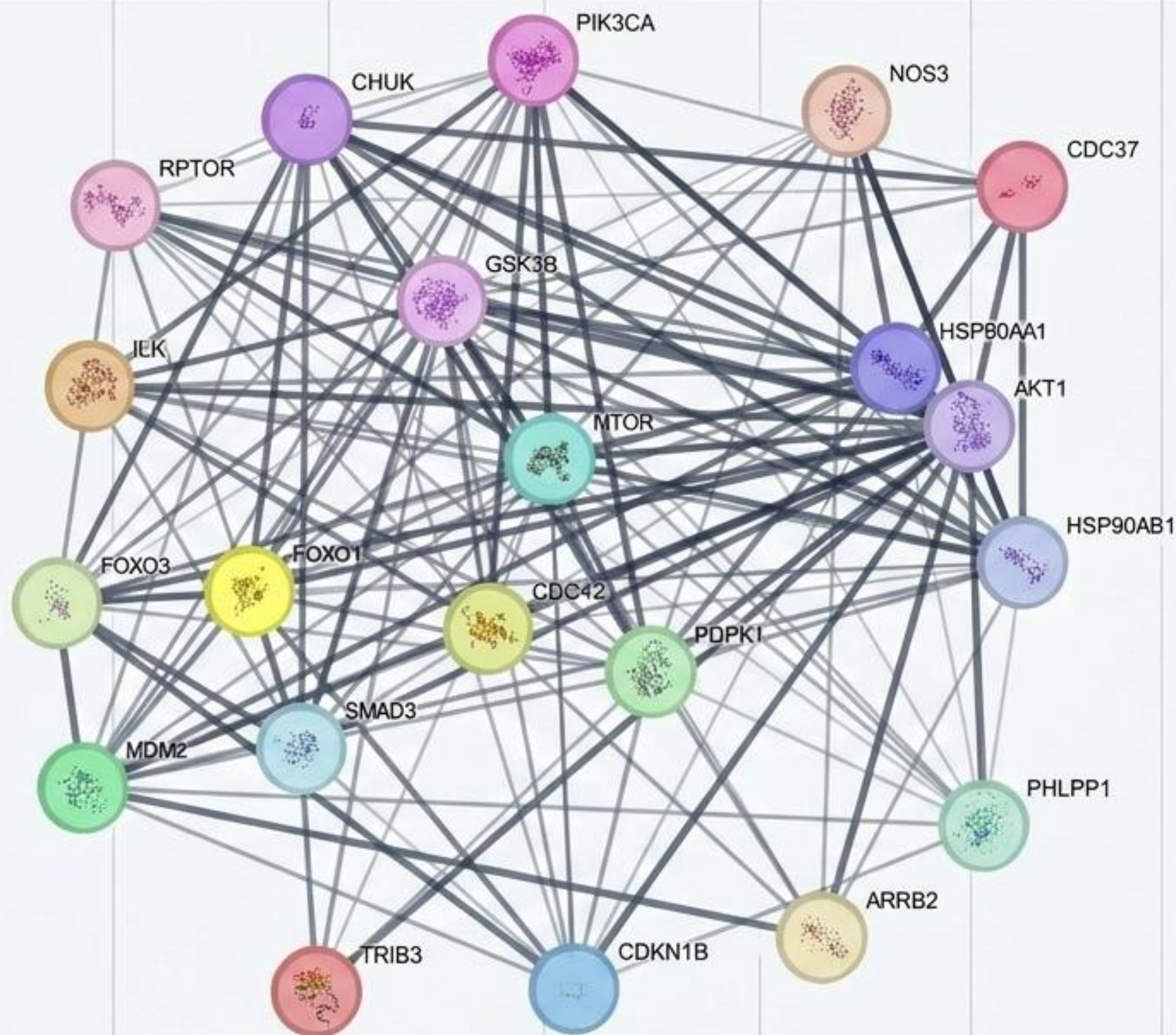
Edges

0.657

Network Density

0.785

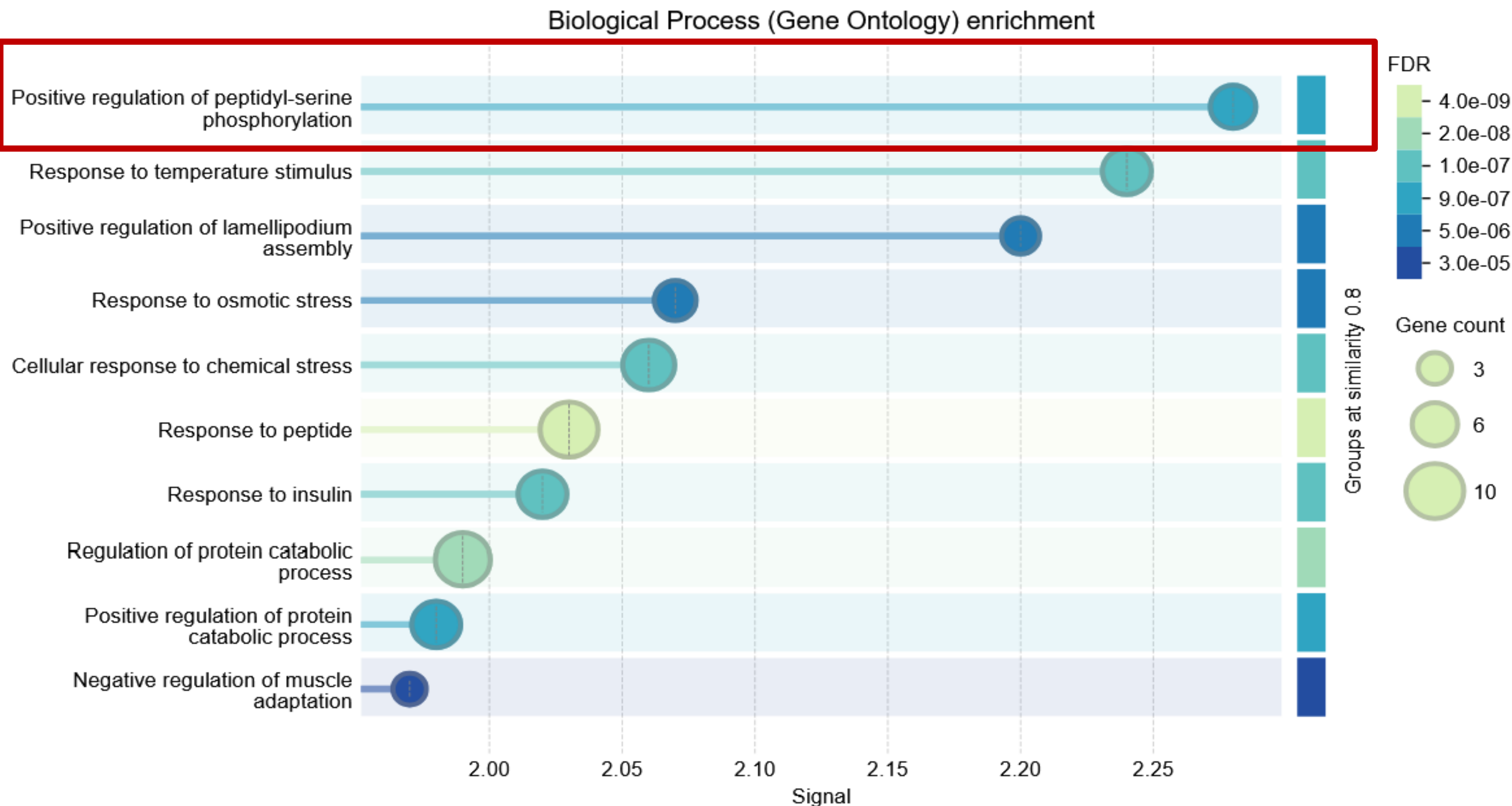
Clustering Coefficient



Avg Neighbors: 13.14 | Diameter: 2 | Radius: 1 | Path Length: 1.34 | Centralization: 0.379 | Time: 0.022s

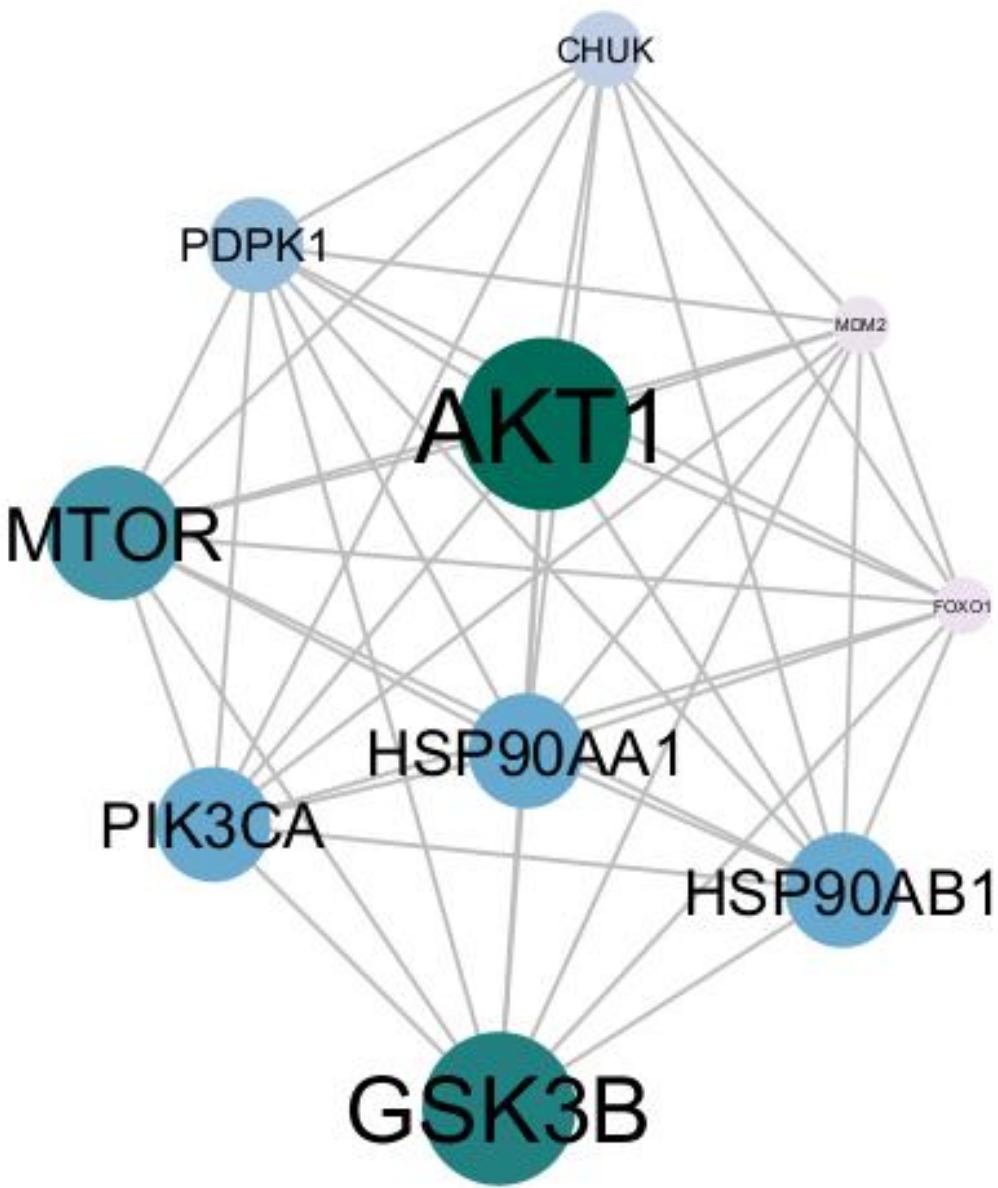
Biological Process

Fungsi AKT1 sebagai *serine/threonine kinase* yang bertugas memfosforilasi protein targetnya.

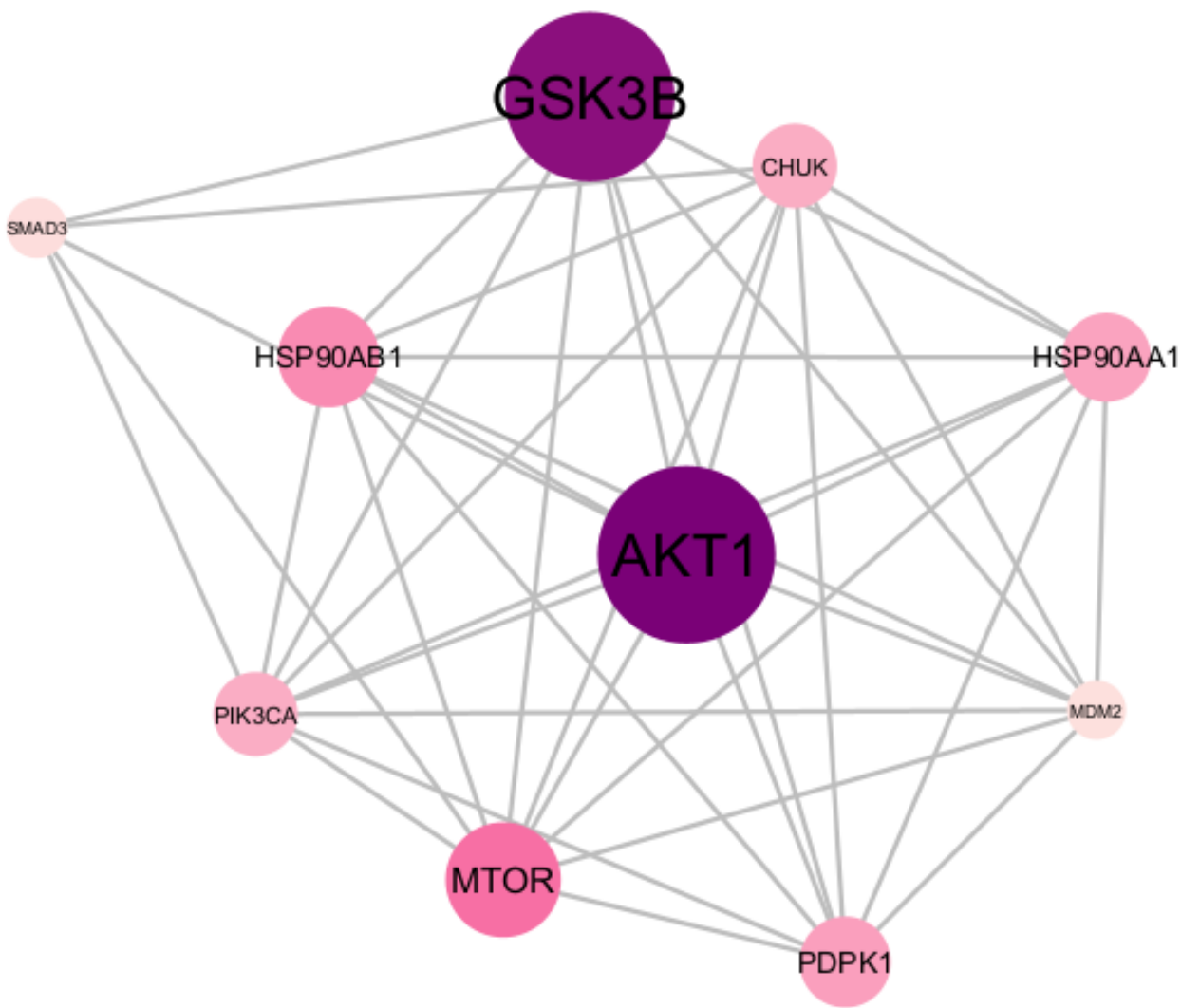


Analisis Topologi

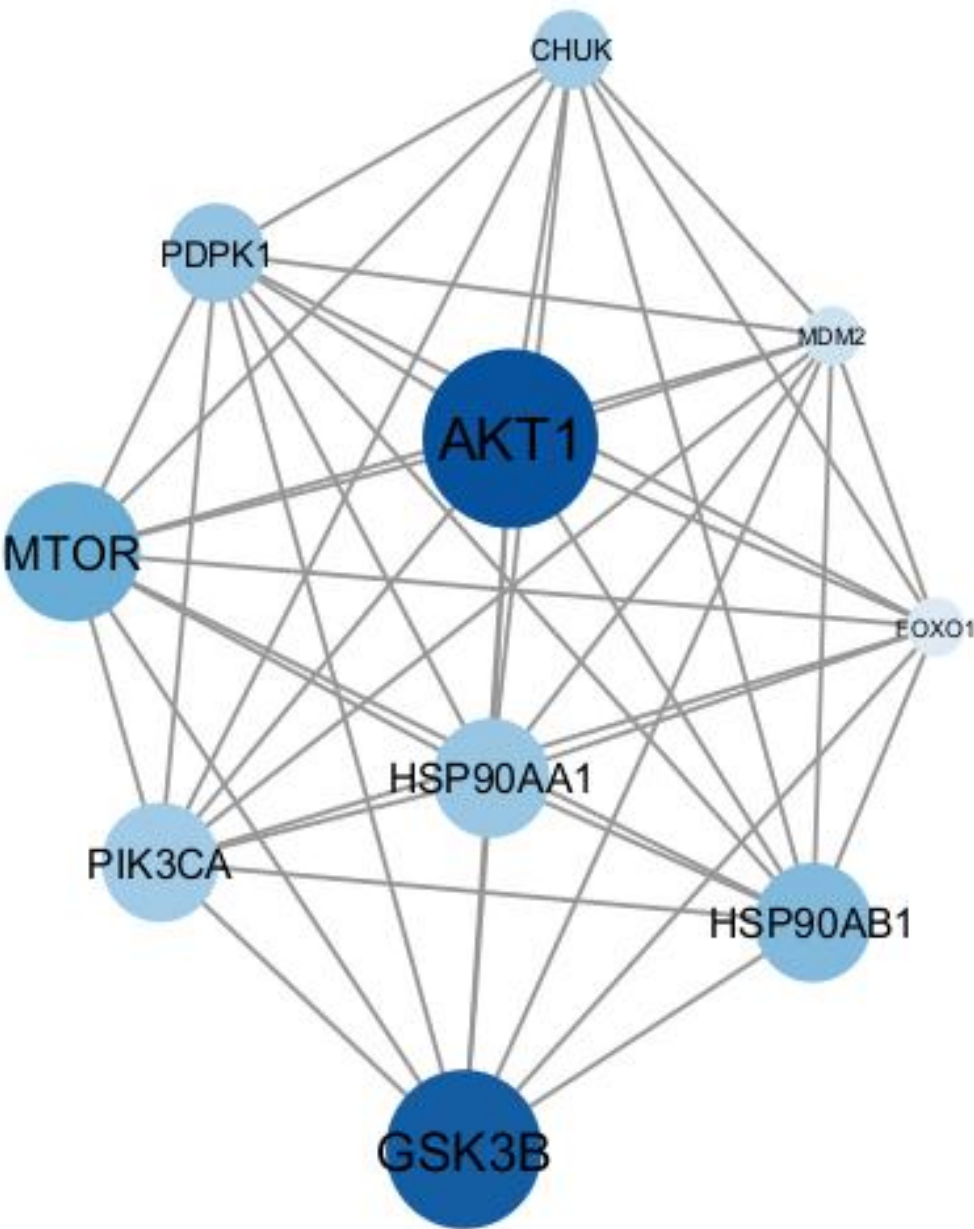
Degree



Betweenness



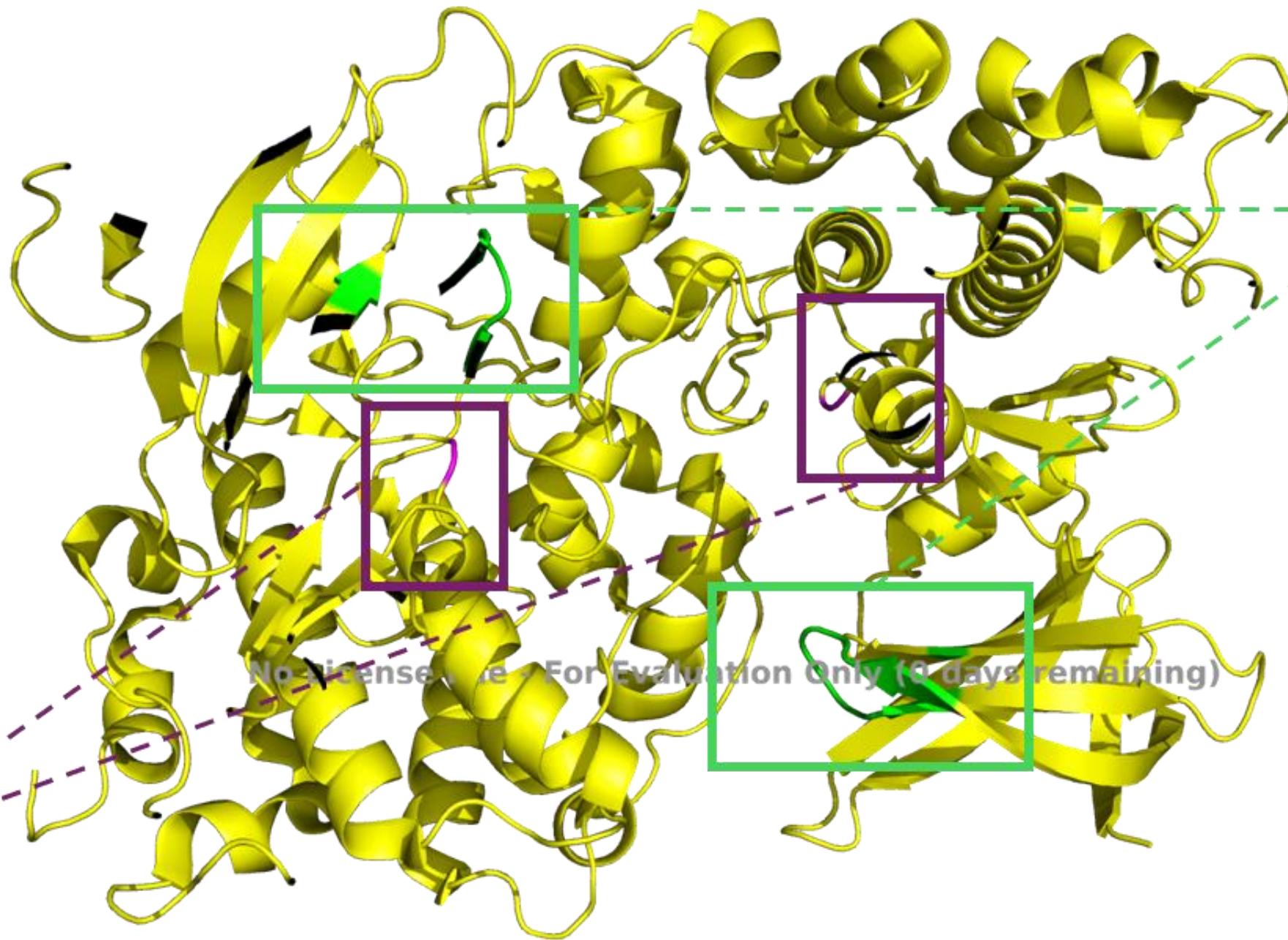
Closeness



Network Analysis Table Top 10 Protein

AKT1			Degree: 20	Closeness: 20.0	Betweenness: 25.01
GSK3B			Degree: 19	Closeness: 19.5	Betweenness: 23.62
MTOR			Degree: 18		
HSP90AB1			Degree: 17		
HSP90AA1			Degree: 17		
PIK3CA			Degree: 17		
PDPK1			Degree: 16		
CHUK			Degree: 15		
MDM2			Degree: 14		

Binding Sites dan Active Sites Identification



Binding site
(residu 62-70 & 85)

Active site
(residu 181)

Structure Stats

Parameter Struktur
(PDB: 9X2Y)

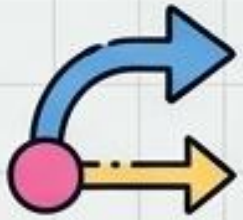
Resolusi	1.96 Å
Rfree	0.25
Clashscore	3
Ramachandran Outliers	0.4%

Pembahasan



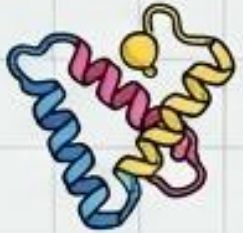
Analisis Jaringan Protein

Jaringan STRING (21 node, 138 edges) mengonfirmasi AKT1 (degree 20) dan GSK3B (degree 19) sebagai hub utama pengatur sinyal pertumbuhan.



Mekanisme Regulasi

Pengayaan fungsi menunjukkan dominasi positive regulation of peptidyl-serine phosphorylation, di mana AKT1 menghambat GSK3B melalui fosforilasi pada residu Serine 9 (Ser9).



Struktur 3D & Situs Fungsional

Pemetaan struktur menunjukkan pemisahan antara area regulasi (Ser9), situs pengikatan ATP (residu 62–70 dan 85), serta situs aktif katalitik (residu 181).



Validasi Komputasional

Analisis PrankWeb mengonfirmasi keandalan geometris struktur melalui Pocket 1 berprobabilitas tinggi (0.937) yang mencakup area katalitik dan pengikatan.



Implikasi Kanker

Hiperaktivitas AKT1 pada oral carcinoma menyebabkan fosforilasi konstan di Ser9 yang menonaktifkan fungsi pro-apoptosis GSK3B, menjadikannya poros strategis untuk target terapi inhibitor.

Keterbatasan Penelitian

Pendekatan In Silico

Penelitian murni berbasis komputasional.
Membutuhkan uji laboratorium empiris (wet-lab) untuk validasi fisik.

Dinamika Konformasi

Keterbatasan model struktural statis dalam menangkap fleksibilitas fleksibilitas penuh wilayah tak beraturan (disordered regions) pada residu target.

Cakupan Jaringan

Dibatasi oleh parameter database saat ini; belum memetakan keseluruhan kompleksitas kompleksitas mikroekosistem tumor secara komprehensif.

Kesimpulan

1.

Hubungan Jaringan Solid

Jalur AKT1-GSK3B bertindak sebagai pengatur utama (Degree 20 & 19).

2.

Mekanisme Kunci

Positive regulation of peptidyl-serine phosphorylation mendorong inaktivasi penekan tumor.

3.

Relevansi Klinis

Disregulasi poros ini secara langsung mengontrol proliferasi dan kelangsungan hidup sel karsinoma rongga mulut.

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